

Elpro International School, Chinchwad
IB Diploma Programme 2024-25

CANDIDATE
NAME

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CENTRE NUMBER

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CANDIDATE NUMBER

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Summative Assessment 2

GROUP 4 – Chemistry- (SL)

GRADE- IBDP Year 1

Paper 2

Friday, 25th April 2025

1 hour 30 minutes

Candidates answer on the same sheet

READ THESE INSTRUCTIONS FIRST

- Write your candidate number and name on all the work you hand in.
 - Write in dark blue or black ball pen.
 - Answer all questions.
 - You should pay attention to punctuation, spelling, and handwriting.
 - At the end of the examination fasten all your work securely together.
 - The marks are given in brackets [] at the end of each question or part question.
 - A calculator and databooklet are required for this paper.
 - The Maximum marks for paper 2 is [50 Marks].
 - Total number of printed pages is 17.
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Assessment Objectives:

- Demonstrate knowledge and understanding of facts, concepts, and terminology.
- Apply facts, concepts, and terminology.
- Formulate, analyse and evaluate scientific explanations.

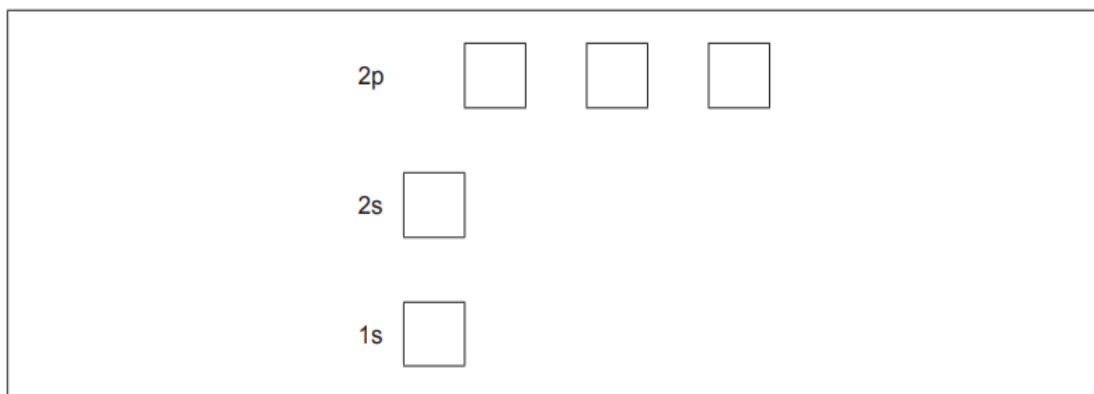
Assessment criteria:

Criteria	Description	Scores attained	Max Marks
Criterion A	Displays subject knowledge		15
Criterion B	Applies relevant information, concepts, and principles		15
Criterion C	Formulate, analyse and evaluate scientific explanations		20
Total			50

Q1 Ammonia, NH_3 , is industrially important for the manufacture of fertilizers, explosives and plastics.

(a) Draw arrows in the boxes to represent the electron configuration of a nitrogen atom.

[2]



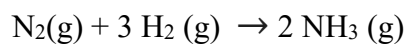
(b) Draw the Lewis (electron dot) structure of the ammonia molecule.

[1]



(c) Ammonia is produced by the Haber–Bosch process which involves the equilibrium:

[1]



(i) Deduce the expression for the equilibrium constant, K_c , for this equation.

(ii) Explain why an increase in pressure shifts the position of equilibrium towards the products and how this affects the value of the equilibrium constant, K_c .

[2]

(iii) State how the use of a catalyst affects the position of the equilibrium.

[1]

Q2. Urea, $(\text{H}_2\text{N})_2\text{CO}$, is excreted by mammals and can be used as a fertilizer.

(a) Calculate the percentage by mass of nitrogen in urea to two decimal places using section 6 of the data booklet.

[2]

(b) (i) Suggest one reason why urea is a solid and ammonia a gas at room temperature.

[1]

- (c) The combustion of urea produces water, carbon dioxide and nitrogen.
Formulate a balanced equation for the reaction.

[2]

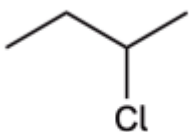
Q3. Alkanes form a homologous series.

- (a) (i) Outline the meaning of homologous series.

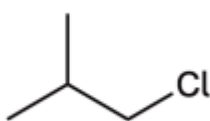
[1]

(ii) State the preferred IUPAC name for the following compounds.

[2]



IUPAC name :



IUPAC name :

(iii) But-2-ene can be polymerized. Draw a section of the resulting polymer showing two repeating units.

[1]

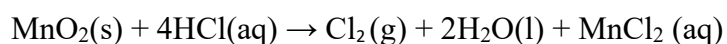


Q4. (a) Chlorine undergoes many reactions.

(i) State, giving a reason, whether the chlorine atom or the chloride ion has a larger radius.

[1]

(b) 2.67g of manganese (IV) oxide was added to 200.0cm³ of 2.00mol dm⁻³ HCl.



i) Calculate the amount, in mol, of manganese (IV) oxide added.

[1]

ii) Determine the limiting reactant, showing your calculations.

[2]

iii) Determine the excess amount, in mol, of the other reactant.

[1]

Q5. Boron is most often encountered as a component in borosilicate glass (heat resistant glass). The naturally occurring element contains two stable isotopes, ${}^5\text{B}^{10}$ and ${}^5\text{B}^{11}$.

(a) State the number of protons, neutrons and electrons in an atom of ${}^5\text{B}^{11}$.

[1]

	Protons	Neutrons	Electrons
${}^{11}_5\text{B}$			

(b) The relative atomic mass of boron is 10.8, to three significant figures. Calculate the percentage of ${}^5\text{B}^{10}$ in the naturally occurring element.

[2]

(c) (i) State the formula of the compound that boron forms with fluorine.

[1]

(ii) Does boron fulfil octet in the (c) (i) compound? Explain.

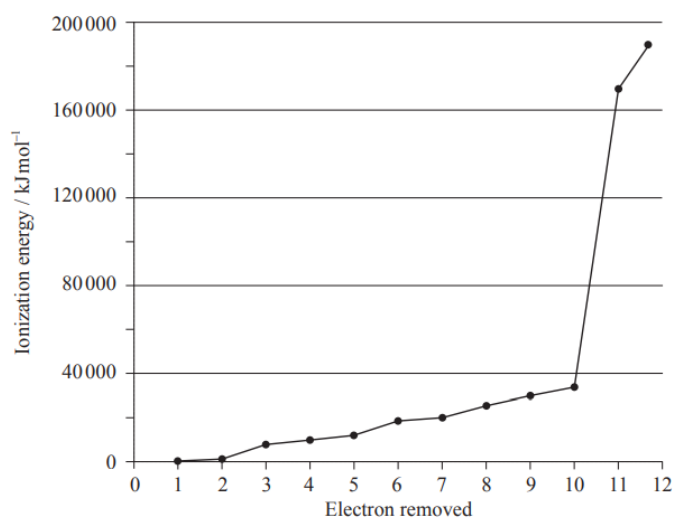
[3]

Q6. A sample of magnesium contains three isotopes: magnesium-24, magnesium-25 and magnesium-26, with abundances of 77.44%, 10.00% and 12.56% respectively.

(a) (i) Calculate the relative atomic mass of this sample of magnesium correct to two decimal places. [2]

(ii) Predict the relative atomic radii of the three magnesium isotopes, giving your reasons. [2]

(b) A graph of the successive ionization energies of magnesium is shown below.



(i) Explain the increase in ionization energy values from the 3rd to the 8th electrons.

[1]

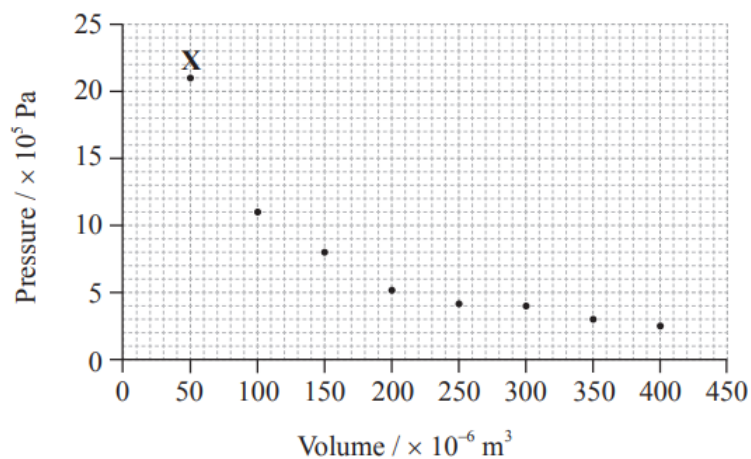
(ii) Explain the sharp increase in ionization energy values between the 10th and 11th electrons. [2]

(c) (i) Magnesium reacts with oxygen to form an ionic compound, magnesium oxide. Describe how the ions are formed, and the structure and bonding in magnesium oxide. [2]

(ii) Carbon reacts with oxygen to form a covalent compound, carbon dioxide. Describe what is meant by a covalent bond. [1]

(d) Define the term electronegativity. [1]

(e) The graph below shows pressure and volume data collected for a sample of carbon dioxide gas at 330 K.



Draw a best-fit curve for the data on the graph.

[1]

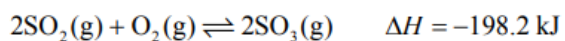
Q7. Chemical equilibrium and kinetics are important concepts in chemistry.

(a) A glass container is half-filled with liquid bromine and then sealed. The system eventually reaches a dynamic equilibrium. State one characteristic of a system in equilibrium.

[1]

(b) The oxidation of sulfur dioxide is an important reaction in the Contact process used to manufacture sulfuric acid.

[1]

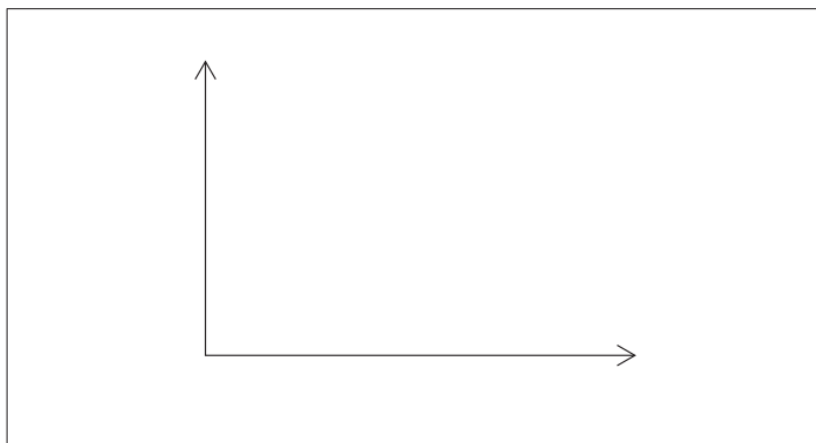


(i) Deduce the equilibrium constant expression, K_c .

(c) Vanadium(V) oxide, V_2O_5 , is a catalyst that can be used in the Contact process. It provides an alternative pathway for the reaction, lowering the activation energy, E_a .

[3]

Sketch the two Maxwell–Boltzmann energy distribution curves for a fixed amount of gas at two different temperatures, T_1 and T_2 ($T_2 > T_1$). Label both axes.

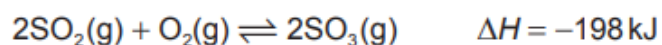


Q8. The rate of reaction is an important factor in industrial processes such as the Contact process to make sulphur trioxide, SO_3 (g).

(a) Define the term rate of reaction.

[1]

(b) The Contact process involves this homogeneous equilibrium:



(i) State and explain how increasing the pressure of the reaction mixture affects the yield of SO_3 .

[2]

(ii) State the effect of increasing temperature on the value of K_c for this reaction.

[2]

Prepared By:		
Reviewed By:		
Approved By:		